

# Presenter Direction & Cheat Sheet

How to present the InsureX RAG sales assistant on the **real product documents**: the running order, what to show, what to say, the live demo script, and one-page facts and answers to keep next to you.

91%

overall accuracy on the 100-question test (91/100)

91%

answer accuracy, 85 answerable questions

100%

right document in the top 4 results

93%

15 out-of-scope questions refused

All numbers in this document come from the latest runs (2026-10-05): `python main.py eval`, `python main.py demo` and `pytest`, with `qwen3:14b + bge-m3` on an RTX 4070 SUPER.

## What the panel grades

The brief grades four things. Tie every part of the talk back to one of them.

| Criterion      | Your evidence                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------|
| RAG precision  | 100-question test on the real PDFs, citations on every fact, grader + retry + refusal path        |
| Workflow logic | 14-node LangGraph: conditional routing, a retrieval cycle, typed state, checkpointed per session  |
| Data integrity | Leads validated inside the MCP tool (Pydantic): Thai phone format, positive income, trimmed names |
| Code quality   | Small single-purpose classes, one Settings object, all prompts in one file, 36 automated tests    |

## Before you start (10 minutes before)

- ☐ Ollama is running and both models are pulled: `ollama list` shows `qwen3:14b` and `bge-m3`.
- ☐ Double-click `start_chatbot.bat` (or `python main.py serve`) and open `http://127.0.0.1:8000`.
- ☐ Send one warm-up question and delete that chat. The first answer loads the model (about 30 s); after that answers take a few seconds.
- ☐ Open these in tabs: `logs/eval_report.md`, the folder `knowledge_base/` with one PDF open, and VS Code on `src/sales_agent/agent.py` (method `build`), `knowledge_base.py` (`PdfLoader`) and `prompts.py`.
- ☐ Keep this guide on a second screen or printed. Close notifications.

## Running order (about 17 minutes + Q&A)

| # | Part                                  | Show on screen                                  | Min |
|---|---------------------------------------|-------------------------------------------------|-----|
| 1 | Opening: the story in 60 seconds      | Web chat, empty                                 | 1   |
| 2 | The brief and how it is covered       | README table                                    | 1.5 |
| 3 | The real data and how it is read      | The PDF folder, one PDF                         | 2.5 |
| 4 | Architecture and the LangGraph flow   | Diagram on the next page, <code>agent.py</code> | 3   |
| 5 | Live demo (includes both bonus tasks) | Web chat                                        | 5   |
| 6 | Results of the 100-question test      | <code>logs/eval_report.md</code>                | 2   |
| 7 | Lessons, next steps, close            | Web chat or VS Code                             | 2   |

1

Opening: the story in 60 seconds

~1 min

**Say:** *I built a sales assistant for InsureX that answers only from the company's product documents and cites the page for every fact. It runs on InsureX's own catalogue: five PDFs built from `insurex.co.th`, one per product category (health, accident, savings, travel and pet), mostly in Thai. It's a LangGraph state machine: it routes each message, searches a FAISS index, checks that the passages really answer the question, retries once with a better query, and if there's still nothing it says so instead of guessing. When a customer is interested it collects their details and saves a validated lead through an MCP tool, and every chat has its own memory. Everything runs locally. On a 100-question test it scored 91% overall.*

2

The brief and how it is covered

~1.5 min

- **Framework:** LangChain for models, prompts and splitting; LangGraph for the workflow (state, nodes, conditional edges, a cycle).
- **Vector DB:** FAISS, chosen over ChromaDB because it is an in-process library with no server, saved as one file. It is wrapped in my own `FaissVectorStore` class, so switching is one class.
- **Not-found handling:** similarity floor → LLM grader → query rewrite → retry → polite refusal. It never invents an answer.
- **Bonus 1** (lead collection via MCP) and **bonus 2** (session memory) are both done; you show them in the demo.

**Say:** *Here's the brief and how I covered it. Everything required is there and I did both bonus tasks. I'll go deeper on three things: how it avoids wrong answers, how the real Thai documents are read, and how leads and memory work.*

**Show** the folder `knowledge_base/`: 5 PDFs → 152 chunks (1000 characters, 150 overlap), embedded with `bge-m3` (multilingual, so English questions find Thai text).

| Document                                    | What it is                                                                                                                                                                        |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>InsureX_Health_Insurance.pdf</code>   | 7 health products: Easy E-Health, E-Health Mini, HB Plus, คั้มตลอดชีพ พลัส, คั้มรักษาเหมาจ่าย เอ็กซ์ตรา (+ for kids), Prima Care; benefit tables with up to 10 plans side by side |
| <code>InsureX_Accident_Insurance.pdf</code> | 38 personal accident plans from 4 insurers (ทิพย, ไทยวิวัฒน์, วัริยะ, เทเวศ) for kids, working age and seniors, with a coverage card per plan                                     |
| <code>InsureX_Savings_Insurance.pdf</code>  | 3 savings plans: Easy E-Save 10/3, คั้มมั่งมี 18/9, คั้มออมสุข                                                                                                                    |
| <code>InsureX_Travel_Insurance.pdf</code>   | Travel insurance: what it covers, optional extras and 28 FAQ answers                                                                                                              |
| <code>InsureX_Pet_Insurance.pdf</code>      | TIP Pet Lover: benefit table for plans S–XXL, conditions, exclusions, FAQ                                                                                                         |

#### HOW THE DATA WAS COLLECTED

Each PDF was generated from `insurex.co.th`: the product data embedded in the category page, plus the partner product pages it links to. Several of those pages build their content in the browser (the accident plan comparison, the online health, travel and pet pages), so they were loaded in Edge with Playwright, "show more" was clicked until every plan appeared, and the FAQ answers were expanded one by one.

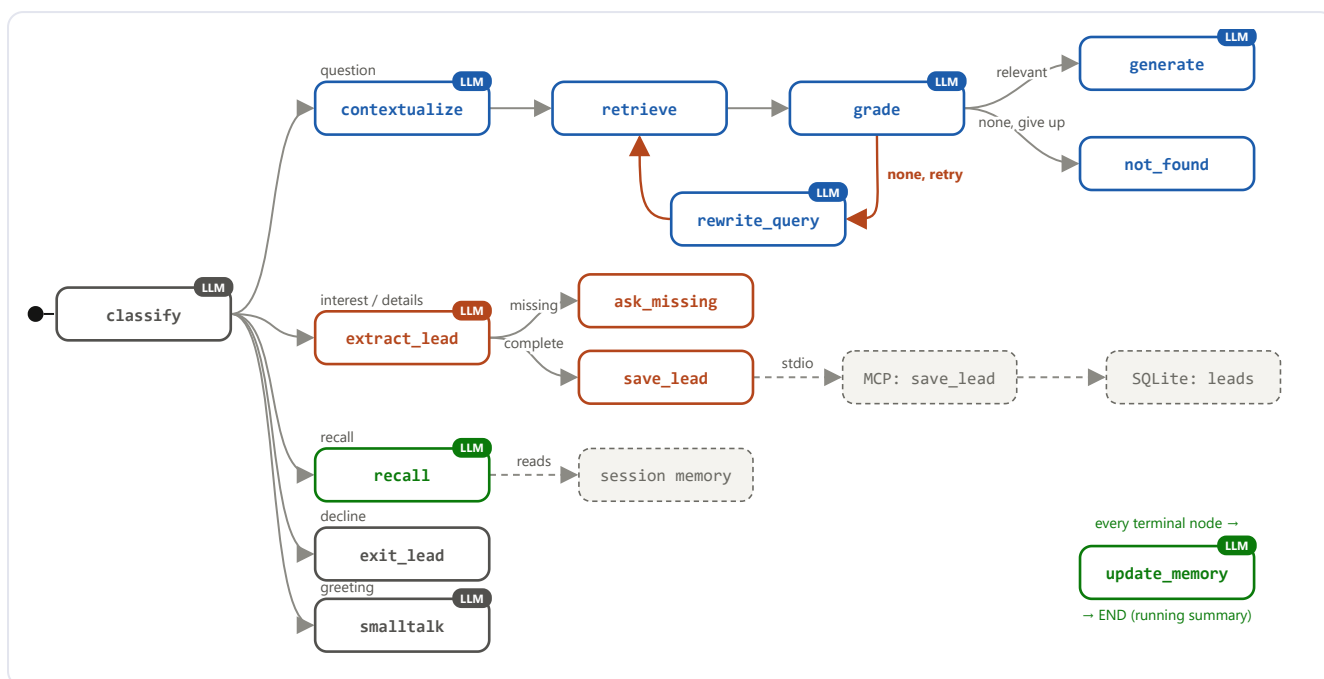
#### THREE READING PROBLEMS YOU FOUND AND FIXED (THIS IS THE STRONGEST PART OF THE STORY)

1. **Thai text was broken.** `pypdf` dropped tone marks and upper vowels ("เบื ย" instead of "เบื๋ย"). Switched the loader to **PyMuPDF**, which reads the text correctly.
2. **Tables fell apart.** Each table cell came out on its own line, so a premium lost its row. The loader now **joins text that shares a baseline**, so "รายปี 36,500 109,500 182,000 362,000" stays one row.
3. **A catalogue mixes many products.** A benefit table rarely repeats its product name, so "แผน M 3,000" could belong to any of seven health products. Every chunk is now labelled "**document title > section**" (e.g. "ประกันสุขภาพ — InsureX > สัญญาเพิ่มเติมคั้มรักษาพริมา แครี"), taken from the headings and carried over to the next pages.

**Say:** *The documents are InsureX's own catalogue, and reading them well took work. The first PDF library lost the Thai tone marks, tables came out one cell per line, and because each PDF holds many products, a table on its own didn't say which product it was for. I fixed all three in the loader and wrote tests for each.*

#### Q: What if a document is a scanned image?

**A:** The loader only reads text layers. I'd add OCR (Tesseract with Thai, or a document-AI service) before chunking; the rest of the pipeline stays the same.



### HOW TO WALK THROUGH IT

1. **classify** (LLM, structured output) picks the intent: question, interest, details, recall, decline or greeting. It sees the mode and recent history.
2. **Question path**: contextualize (rewrites follow-ups using memory, in Thai, with the Thai product names) → retrieve (top 4, floor 0.45) → grade → generate with citations.
3. **The cycle**: if the grader keeps nothing, rewrite\_query → retrieve once more, then not\_found. The retry counter lives in the state, so it can't loop forever.
4. **Lead path**: extract\_lead → ask\_missing (fields missing) or save\_lead (complete, through the MCP server).
5. **Memory**: recall answers questions about the chat itself; every turn ends at update\_memory, which keeps a running summary.

### STATE (AGENTSTATE)

messages (add\_messages reducer) · mode (qa / lead\_collection, persists across turns) · lead (partial draft) · query, attempts, documents (per turn) · summary, summarized\_upto (memory).

**Say:** This is the heart of it. Every message goes to a router first. Questions take the RAG path: rewrite the question using memory, search, grade, answer with citations. The orange loop is the cycle: if nothing relevant comes back, it rephrases and searches once more before refusing. Interest switches the agent into lead mode, which persists in the state. The badges show which steps call the model; the routing itself is plain Python, which I unit-test.

### Q: Why not a ReAct agent that picks its own tools?

**A:** A free-form agent is harder to predict and test. Here the LLM classifies, grades and writes, but the control flow is code, so every turn's path is logged and rules like "never answer without relevant passages" are guaranteed.

### Q: Why a local model?

**A:** Customer income and phone numbers never leave the machine, and there's no per-call cost. The model is one setting (CHAT\_MODEL), so a hosted model is a configuration change.

## 5 Live demo script

~5 min

Type these messages in the web chat, in order. Point at the **source chips** and the **Agent pipeline** panel lighting up each node. The full recorded run is in the table that follows, so you know what each answer should look like.

1 What does Prima Care plan M pay for room and board per day?

English question, Thai catalogue. Expect 3,000 baht per day with a citation to `InsureX_Health_Insurance.pdf`.  
Point at `classify` → `contextualize` → `retrieve` → `grade` → `generate`.

2 And what about the ICU room on the same plan?

**Memory:** "the same plan" is resolved to Prima Care plan M. Expect 6,000 baht.

3 Do you sell car insurance?

**Not-found path:** point at the orange retry (`rewrite_query` → `retrieve`) and the polite refusal. It doesn't invent a product.

4 ประกันอุบัติเหตุ ทิพย์ TIP PA SENIOR แผน 1 เบี้ยเท่าไร

**Finding one plan among 38**, in Thai. Expect 1,200 baht a year.

5 I'm interested in Prima Care. Can an agent contact me?

**Bonus 1:** the lead badge appears and the progress ring starts at 0/4.

6 I'm Mai, a nurse, 30,000 a month, phone 12345

**Data integrity:** the MCP tool rejects the phone; name, job and income are kept. Then send  
`sorry, 081-234-5678` : the lead is saved, a toast appears, and it shows under Saved leads.

7 Click **New chat** and ask What's my name?

**Bonus 2:** the new chat knows nothing about Mai (separate sessions). Go back to the first chat: its history is still there.

8 Hover the first chat and click **delete**. Say that this erases its memory, while the saved lead stays as a business record.

**If something goes wrong live:** say "the model is local, so let me show the recorded run", and open `logs/demo_transcript.md` or the table below. The same script replays with `python main.py demo`. If Ollama stopped, restart it; the chat history survives.

### Recorded demo run (17 turns, 3 leads saved)

| Session | Customer                                                    | Assistant (shortened)                                                                                                                      | Path                                                                                                                                                 | Time |
|---------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| alice   | Hi there!                                                   | Hi there! How can I assist you today? I'm here to help with health, personal accident, savings, travel, and pet insurance through InsureX. | <code>classify</code> → <code>smalltalk</code> → <code>update_memory</code>                                                                          | 1.9s |
| alice   | What does Prima Care plan M pay for room and board per day? | Plan M of Prima Care pays up to 3,000 THB per day for room and board [ <code>InsureX_Health_Insurance.pdf</code> , p.25].                  | <code>classify</code> → <code>contextualize</code> → <code>retrieve</code> → <code>grade</code> → <code>generate</code> → <code>update_memory</code> | 6.8s |
| alice   | And what about the ICU room on the same plan?               | Plan M of Prima Care pays up to 6,000 THB per day for ICU room [ <code>InsureX_Health_Insurance.pdf</code> , p.25].                        | <code>classify</code> → <code>contextualize</code> → <code>retrieve</code> → <code>grade</code> → <code>generate</code> → <code>update_memory</code> | 4.8s |
| alice   | Do you sell car insurance?                                  | Sorry, I couldn't find this in the InsureX product documents, so I can't give a reliable answer. I can help with InsureX                   | <code>classify</code> → <code>contextualize</code> → <code>retrieve</code> → <code>grade</code> →                                                    | 4.5s |

|         |                                                                                  |                                                                                                                                                                                                                                             |                                                                              |       |
|---------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------|
|         |                                                                                  | health, personal accident, savings, travel and pet insurance.                                                                                                                                                                               | rewrite_query → retrieve<br>→ grade → not_found →<br>update_memory           |       |
| alice   | I'm interested in Prima Care. Can an agent contact me?                           | Great to hear you're interested in Prima Care! So an agent can contact you, could you share your full name, your occupation, your monthly income (THB) and a contact phone number?                                                          | classify → extract_lead →<br>ask_missing →<br>update_memory                  | 1.5s  |
| alice   | My name is Alice Wong and I work as a nurse                                      | Thank you Alice Wong! Could you also share your monthly income (THB) and a contact phone number?                                                                                                                                            | classify → extract_lead →<br>ask_missing →<br>update_memory                  | 1.4s  |
| alice   | Before that - up to what age can I renew it?                                     | For Prima Care, you can renew the policy up to age 79, with coverage continuing until age 80 [InsureX_Health_Insurance.pdf, p.26]. To continue with your application, could you share your monthly income (THB) and a contact phone number? | classify → contextualize<br>→ retrieve → grade →<br>generate → update_memory | 11.3s |
| alice   | I earn about 32,000 baht a month, my number is 089-765-4321                      | Thank you, Alice Wong! Your details are saved (lead #1). An InsureX agent will call you on 0897654321 about Prima Care shortly.                                                                                                             | classify → extract_lead →<br>save_lead → update_memory                       | 4.7s  |
| alice   | What's my name, and what did I ask about first?                                  | Your name is Alice Wong, and you first asked about Prima Care plan M's coverage for room and board.                                                                                                                                         | classify → recall →<br>update_memory                                         | 1.5s  |
| somchai | ประกันสัตว์เลี้ยง แผน XL เบี้ยปีละเท่าไร และค่ารักษาจากการเจ็บป่วยต่อครั้งเท่าไร | เบี้ยประกันภัยสำหรับแผน XL คือ 8,400 บาท/ปี [InsureX_Pet_Insurance.pdf, p.3] ค่ารักษาพยาบาลจากการเจ็บป่วยต่อครั้งสำหรับแผน XL คือ 1,500 บาท (สูงสุด 10 ครั้ง/ปี) [InsureX_Pet_Insurance.pdf, p.2]                                           | classify → contextualize<br>→ retrieve → grade →<br>generate → update_memory | 7.5s  |
| somchai | ประกันอุบัติเหตุ ทิพย์ TIP PA SENIOR แผน 1 เบี้ยเท่าไร                           | เบี้ยประกันสำหรับประกันอุบัติเหตุ ทิพย์ TIP PA SENIOR แผน 1 คือ 1,200 บาท/ปี [InsureX_Accident_Insurance.pdf, p.7]                                                                                                                          | classify → contextualize<br>→ retrieve → grade →<br>generate → update_memory | 9.2s  |
| somchai | สนใจรับ ผมชื่อ สมชาย ใจดี เป็นวิศวกร เงินเดือน 5 หมื่น เบอร์ 081-234-5678        | ขอบคุณค่ะ คุณสมชาย ใจดี บันทึกข้อมูลเรียบร้อยแล้ว (หมายเลข 2) ตัวแทนจะติดต่อกลับที่เบอร์ 0812345678 เร็ว ๆ นี้ค่ะ                                                                                                                           | classify → extract_lead →<br>save_lead → update_memory                       | 4.7s  |
| dave    | I'd like to apply for travel insurance                                           | Great to hear you're interested in ประกันการเดินทาง! So an agent can contact you, could you share your full name, your occupation, your monthly income (THB) and a contact phone number?                                                    | classify → extract_lead →<br>ask_missing →<br>update_memory                  | 1.5s  |
| dave    | Dave Miller, software developer, 85000 per month, phone 12345                    | Sorry, some details don't look right. Could you check a contact phone number again?                                                                                                                                                         | classify → extract_lead →<br>save_lead → update_memory                       | 2.8s  |
| dave    | Sorry, it's 091-555-0123                                                         | Thank you, Dave Miller! Your details are saved (lead #3). An InsureX agent will call you on 0915550123 about ประกันการเดินทาง shortly.                                                                                                      | classify → extract_lead →<br>save_lead → update_memory                       | 2.9s  |
| bob     | What's my name? What did I ask you before?                                       | I don't have any previous conversation history to reference.                                                                                                                                                                                | classify → recall →<br>update_memory                                         | 0.9s  |

|       |                                           |                                                                 |                                   |      |
|-------|-------------------------------------------|-----------------------------------------------------------------|-----------------------------------|------|
| alice | Can you remind me what the room rate was? | The room rate for Prima Care plan M is up to 3,000 THB per day. | classify → recall → update_memory | 2.8s |
|-------|-------------------------------------------|-----------------------------------------------------------------|-----------------------------------|------|

Leads saved in that run

| # | Name        | Occupation         | Income | Phone      | Product                 | Session |
|---|-------------|--------------------|--------|------------|-------------------------|---------|
| 1 | Alice Wong  | nurse              | 32,000 | 0897654321 | Prima Care              | alice   |
| 2 | สมชาย ใจดี  | วิศวกร             | 50,000 | 0812345678 | ประกันสัตว์เลี้ยง (pet) | somchai |
| 3 | Dave Miller | software developer | 85,000 | 0915550123 | ประกันการเดินทาง        | dave    |

## 6 Results of the 100-question test

~2 min

**How the test works:** 85 questions with a known answer from the PDFs (Thai and English, every expected value checked against the source text) plus 15 questions about products that aren't in the documents (car, home, pension, cyber, and the two products that were removed from the knowledge base). Each question runs in a fresh session through the full graph. A question passes if the expected fact appears in the answer, or, for out-of-scope questions, if the assistant refuses.

| Document                       | Chunks | Questions | Retrieval | Correct |
|--------------------------------|--------|-----------|-----------|---------|
| InsureX_Health_Insurance.pdf   | 68     | 24        | 100%      | 83%     |
| InsureX_Accident_Insurance.pdf | 49     | 18        | 100%      | 83%     |
| InsureX_Savings_Insurance.pdf  | 14     | 15        | 100%      | 100%    |
| InsureX_Travel_Insurance.pdf   | 14     | 13        | 100%      | 100%    |
| InsureX_Pet_Insurance.pdf      | 7      | 15        | 100%      | 93%     |
| Out of scope (should refuse)   | –      | 15        | –         | 93%     |

### Choosing the chunk size

| Chunk / overlap (characters) | Chunks | Overall | Answers right | Refused correctly |
|------------------------------|--------|---------|---------------|-------------------|
| 500/100                      | 304    | 85/100  | 71/85         | 14/15             |
| 700/120                      | 220    | 82/100  | 69/85         | 13/15             |
| <b>1000/150 (chosen)</b>     | 152    | 91/100  | 77/85         | 14/15             |
| 1200/200                     | 138    | 89/100  | 76/85         | 13/15             |

1000/150 scored best (91/100, 77/85 answers, 14/15 refusals). Smaller chunks scored lower (85 and 82) and 1200/200 scored 89 with one fewer correct refusal. The likely reason is that at about 1000 characters a benefit-table block stays in one chunk with its plan-name headings; this is an explanation, not something the test measured. Overlap stays at 15% so a fact on a chunk boundary is whole in at least one chunk.

### Questions that still fail (9)

| #  | Question                                                                           | Expected | Retr. | Answer given                                                                                                                                                                                                                     |
|----|------------------------------------------------------------------------------------|----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | อีซี่ อี เฮลท์ ฝ่าตัดต่อกระจกมีระยะเวลารอคอยกี่วัน                                 | 120      | ✓     | ขอภัยค่ะ ไม่พบข้อมูลเรื่องนี้ในเอกสารผลิตภัณฑ์ของ InsureX จึงไม่สามารถให้คำตอบที่ถูกต้องได้ กรุณาสอบถามเกี่ยวกับประกันสุขภาพ ประกันอุบัติเหตุ ประกันสะสมทรัพย์ ประกันเดินทาง หรือประกันสัตว์เลี้ยง                               |
| 9  | สัญญาเพิ่มเติม เฮซปี พลัส รับประกันอายุสูงสุดกี่ปี                                 | 70       | ✓     | สัญญาเพิ่มเติม เฮซปี พลัส รับประกันอายุสูงสุดถึงอายุ 85 ปี [InsureX_Health_Insurance.pdf, p.5].                                                                                                                                  |
| 12 | HB Plus 1000 เบี้ยประกันต่อปีสำหรับคนอายุ 30 ปีเท่าไร                              | 2,200    | ✓     | เบี้ยประกันต่อปีสำหรับแผน HB Plus 1000 และผู้เอาประกันภัยอายุ 30 ปีคือ 4,400 บาท [InsureX_Health_Insurance.pdf, p.5].                                                                                                            |
| 20 | คุ้มครองเหมาจ่าย เอ็กซ์ตร้า พอร์ คัดส์ แผน Deductible ช่วยลดเบี้ยได้กี่เปอร์เซ็นต์ | 50%      | ✓     | คุ้มครองเหมาจ่าย เอ็กซ์ตร้า พอร์ คัดส์ แผน Deductible ช่วยลดเบี้ยประกันภัยเมื่อเทียบกับแผนแบบไม่มีความรับผิดชอบส่วนแรก (No Deductible) สำหรับผู้เอาประกันภัยอายุ 5 ปี แผน Deductible 25,000   50,000 มีเบี้ยประกันภัยรายปี 62... |
| 28 | วิริยะ PA กระจกหัก แผน 3 เบี้ยประกันเท่าไร                                         | 5,820    | ✓     | วิริยะ PA กระจกหัก แผน 3 มีเบี้ยประกันเท่ากับ 7,560 บาท [InsureX_Accident_Insurance.pdf, p.8].                                                                                                                                   |
| 30 | วิริยะ PA กระจกหัก แผน 1 ค่ารักษากรณีกระจกหักเท่าไร                                | 100,000  | ✓     | ค่ารักษากรณีกระจกหักสำหรับแผน PA กระจกหัก แผน 1 คือ 3,240 บาท [InsureX_Accident_Insurance.pdf, p.15].                                                                                                                            |

|    |                                                                    |         |   |                                                                                                                                                                                                                           |
|----|--------------------------------------------------------------------|---------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37 | PA Life Span ร้อยทำงาน<br>แผน 3 ชดเชยรายได้จาก<br>อุบัติเหตุเท่าไร | 3,000   | ✓ | PA Life Span ร้อยทำงาน แผน 3 ชดเชยรายได้จากอุบัติเหตุสูงสุด 1,000,000 บาท [InsureX_Accident_Insurance.pdf, p.12].                                                                                                         |
| 82 | ประกันสัตว์เลี้ยงรับประกันโดย<br>บริษัทอะไร                        | ทิพย์   | ✓ | ประกันสัตว์เลี้ยงของ InsureX รับประกันโดยบริษัท อินซัวร์ เอกซ์ จำกัด [InsureX_Pet_Insurance.pdf, p.1]                                                                                                                     |
| 91 | ประกันบำนาญ คำนวณได้<br>รับเงินบำนาญปีละกี่เปอร์เซ็นต์             | refusal | – | The context provided does not mention "ประกันบำนาญ คำนวณ" or any details about pension insurance or annual pension percentages. Therefore, I cannot provide the requested information [InsureX_Savings_Insurance.pdf, ... |

**Say:** I tested it on 100 questions written from the real documents, about a third in English and the rest in Thai, plus 15 it should refuse. It got 91% right overall. Every question that fails is listed, along with what I'd do next.

## 7 Lessons, next steps, close

~2 min

### LESSONS (SHOW ENGINEERING JUDGEMENT)

- **Similarity is not relevance.** Off-topic questions still get high cosine scores, so a grader and a retry cycle are needed before answering.
- **Real documents break assumptions.** Broken Thai extraction, split tables and legacy font encodings were only found by testing on the real PDFs.
- **Chunks need context.** A benefit table that never names its product can be mixed up with another product's table. Labelling every chunk with its document and section fixed that class of error.
- **Tune with the test set, not by rule of thumb.** Chunk size was picked by running all 100 questions on several settings: 500/100 → 85, 700/120 → 82, 1000/150 → 91, 1200/200 → 89 (see logs/chunk\_comparison.md).
- **Measure, then change.** Every fix was checked against the same 100 questions, so improvements are numbers, not impressions.

### NEXT STEPS

- Build the test set with the sales team, and add LLM-as-judge scoring for answer quality.
- Hybrid search (BM25 + vectors) and a reranker, which help with exact numbers and plan names.
- Parse tables into structured rows (or a small SQL table) for premium lookups.
- Log-in per agent, PDPA consent before collecting details, encrypted leads, CRM integration through the same MCP tool, Docker deployment.

**Say:** To sum up: a LangGraph agent that answers only from InsureX's documents and refuses when it can't, reads real Thai PDFs correctly, captures validated leads through MCP, remembers each conversation, and runs entirely on the local machine. I'm happy to go into the code or try any question you like.

Numbers

|                |                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Models         | qwen3:14b (chat, tool calling, thinking off) + bge-m3 (embeddings, 1024-d, multilingual) via Ollama                                                 |
| Knowledge base | 5 real PDFs → 152 chunks (1000 chars, 150 overlap), each labelled with its document and section                                                     |
| PDF reading    | PyMuPDF; rows joined by baseline (3 pt tolerance); section = latest heading ≥ 60% of the title size; legacy Thai PUA glyphs mapped to standard Thai |
| Retrieval      | FAISS inner product on L2-normalised vectors (= cosine), top 4, floor 0.45, LLM grader, max 1 rewrite                                               |
| Graph          | 14 nodes, one cycle (rewrite_query → retrieve), recursion limit 25                                                                                  |
| Memory         | SQLite checkpointer, thread = session; last 8 messages + running summary after 12 (keep 6)                                                          |
| Test           | 100 questions: overall 91% · answers 91% · retrieval 100% · refusals 93%                                                                            |
| Demo           | 17 turns, 4 customers + a restart, 3 leads saved; RAG answers ≈ 7.9 s                                                                               |
| Tests          | 36 pytest tests (MCP over real stdio, web API, PDF reading, end-to-end when Ollama is up)                                                           |

Latency by turn type (demo run)

| Turn type              | Turns | Average | Range      |
|------------------------|-------|---------|------------|
| Memory / small talk    | 4     | 1.8 s   | 0.9–2.8 s  |
| RAG answer             | 5     | 7.9 s   | 4.8–11.3 s |
| Not found (with retry) | 1     | 4.5 s   | 4.5–4.5 s  |
| Lead question          | 3     | 1.5 s   | 1.4–1.5 s  |
| Lead saved / validated | 4     | 3.8 s   | 2.8–4.7 s  |

Product facts (in case the panel asks)

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Health<br>7 products             | <b>Easy E-Health:</b> lump-sum IPD cover up to 1.5 million/year, 10% discount next year if no claim, waiting 30 days (120 for listed conditions). <b>E-Health Mini:</b> up to 30,000 per visit and 300,000 per year, waiting 30 days (invasive cancer 90). <b>HB Plus:</b> daily cash 500–10,000, age 1 month 1 day – 70, renew to 84. <b>คุ้มครองชีพ พลัส:</b> age 16–60, plans of 1–12 million/year + 1.5 million on 50 critical illnesses + 300,000 life cover. <b>คุ้มครองสุขภาพสบาย เอ็กซ์ตร้า:</b> Bronze 1 million to Diamond 100 million/year, age 11–70; <b>for kids:</b> age 1 month 1 day – 10, Deductible plan cuts premium 50%. <b>Prima Care:</b> plans S–XXL, room 1,500–7,500/day, renew to 79. Health premiums: tax deduction up to 25,000. |
| Accident<br>38 plans, 4 insurers | <b>เทเวศ "วันละ 1 บาท / 2 บาท":</b> 60–450 baht for 60–365 days, 60,000 death cover (age 20–60). <b>วิริยะ PA กระตุกหัก</b> plans 1–4: 3,240–6,720 baht, death 300,000–1,000,000, bone fracture 100,000–150,000 (age 0–65; seniors 66–70 pay 4,260–8,700). <b>ไทยวิวัฒน์ PA Life Span</b> (school 7–20, working 21–50, seniors 51–65): 3,950–29,900 baht, with 4 popular diseases. <b>ทิพย์ TIP PA SENIOR</b> (55–80): 1,200–6,000 baht.                                                                                                                                                                                                                                                                                                                     |
| Savings<br>3 plans               | <b>Easy E-Save 10/3:</b> pay 3 / cover 10, 30,000–350,000 a year, 303% at maturity, IRR 1.24%, no health questions. <b>คุ้มมั่งมี 18/9:</b> pay 9 / cover 18, minimum 100,000, 379% back in total, age 30 SA 100,000 = 36,500 a year. <b>คุ้มออมสุข:</b> pay 15 / cover 25, minimum 150,000, 150% at maturity, age 30 SA 100,000 = 878 a month (male).                                                                                                                                                                                                                                                                                                                                                                                                       |

|                              |                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Travel</b>                | Domestic or abroad; medical (OPD/IPD), delays, cancellation and luggage; optional home-theft cover, and food poisoning cover on domestic plans. No waiting period, no health check; pre-existing conditions not covered. Schengen visa: at least 30,000 euro medical cover. Price depends on destination, dates and travellers. Call 1314 (9.00–19.00). |
| <b>Pet</b><br>ทิพย์ประกันภัย | Dogs and cats, age 3 months – 9 years (health check needed at 8–9), no microchip needed, policy by email within 30 minutes. Plans S–XXL: 1,950 / 2,950 / 5,400 / 8,400 / 12,000 baht a year, up to 118,000 cover. Illness in the first 60 days and anything outside Thailand are not covered.                                                           |

## Tough questions

### Q: How do you know the answer is grounded and not right by luck?

A: Citations are required after each fact, the grader filters passages, and the test checks retrieval and the answer separately. Next I'd add an automatic "is every claim supported?" check.

### Q: Why 1000 characters with 150 overlap?

A: I tested it on these documents. A fast retrieval sweep over 13 settings picked the candidates, then each ran the full 100-question test: 500/100 → 85, 700/120 → 82, 1000/150 → 91, 1200/200 → 89. 1000/150 scored best (91/100, 77/85 answers, 14/15 refusals). Smaller chunks scored lower (85 and 82) and 1200/200 scored 89 with one fewer correct refusal. The likely reason is that at about 1000 characters a benefit-table block stays in one chunk with its plan-name headings; this is an explanation, not something the test measured. Overlap stays at 15% so a fact on a chunk boundary is whole in at least one chunk.

### Q: Is 100 questions enough?

A: It covers every document, both languages, tables and refusals, and every expected answer was checked against the source text. For production I'd grow it with the sales team's real questions and track it on every change.

### Q: Why not just put the whole document in the prompt?

A: Five documents would fit today, but not a real product catalogue. Retrieval keeps prompts small and fast, and citations show exactly where each fact came from.

### Q: The accident catalogue has 38 plans. Is RAG the right tool for comparing them?

A: It works for single lookups ("what does plan 3 cost?"). For "cheapest plan with fracture cover for a 70-year-old" the better design is to load the plan table into a small database and let the agent query it, keeping RAG for the text.

### Q: Where did the knowledge base come from?

A: The brief's PDFs weren't provided, so I built one PDF per product category from insurex.co.th. Pages that load their data in the browser were rendered in Edge with Playwright, and every plan count was checked against the site's own totals.

### Q: Can the agent save a lead with wrong data?

A: It only extracts what the customer typed (the prompt forbids guessing), and the MCP tool rejects invalid phones, empty names and non-positive income. A human agent calling back is the final check.

### Q: What about prompt injection inside a document?

A: The documents are company-controlled; the answer prompt treats context as facts only; the graph, not the LLM, decides the flow; and the only write action uses validated fields.

### Q: How would it scale?

A: FAISS handles millions of vectors (IVF/HNSW index); move the checkpoint to Postgres, run several API workers, serve the model with vLLM or a hosted endpoint. The graph code doesn't change.

### Q: What happens if Ollama crashes mid-chat?

A: The service catches it, returns "the model service is not reachable", logs it, and the session carries on when Ollama is back.

**Q: PDPA?**

**A:** In production: a consent line before collecting details, encrypted leads, retention rules. Deleting a chat already erases its conversation memory.

**Glossary**

|                                 |                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>RAG</b>                      | Retrieval-Augmented Generation: find relevant passages first, then let the LLM answer only from them.                    |
| <b>Embedding / cosine</b>       | A vector that represents meaning; cosine similarity measures how close two meanings are (1 = identical).                 |
| <b>Chunk</b>                    | A piece of a document (about 1000 characters) that is embedded and retrieved.                                            |
| <b>Grader</b>                   | An LLM step that keeps only passages that directly answer the question.                                                  |
| <b>Conditional edge / cycle</b> | A transition chosen from the state; a cycle goes back to an earlier node (the retry).                                    |
| <b>Checkpoint / thread</b>      | LangGraph saves the state after each step under a thread ID; one thread = one conversation.                              |
| <b>MCP</b>                      | Model Context Protocol: an open standard for exposing tools to AI apps (here over stdio).                                |
| <b>PUA</b>                      | Unicode Private Use Area: code points a font can use for its own glyphs; old Thai fonts put positioned tone marks there. |
| <b>ငါးသောင်း</b>                | Lump-sum health cover: one yearly limit for all hospital costs instead of separate limits per item.                      |
| <b>Deductible</b>               | The part of a claim the customer pays first; a deductible plan has a lower premium.                                      |